Chapter 5 Critical Thinking Questions (page 118: #1 - #9 and #11):

1.

A loop structure allows a segment of code to run multiple times for as long as criteria is met. This increases coding efficiency and organization.

2.

A while loop only executes if the condition at the beginning is met. Meaning if the condition is not met, the loop won't run at all. A do-while statement will always run at least once because the condition is evaluated after the code is run. If the condition is true, then it will continue running.

3.

I believe an input validation loop was required for the review AcountSetup.

4.

a) An infinite loop is a segment of code that will repeatedly run forever because the condition meant to terminate it is never evaluated as false.
b) Logic errors (missing update or control variable mistake) can make it so a loop fails to update or change and the condition will never end up evaluating to false. Boundary condition errors can also make it so the wrong comparison operator is used in the condition. This can mean the condition is never met because the target is skipped over or moved away from.
c) In terms of an infinite loop, overflow means that method calls kept stacking until they eventually exceeded java's allocated stack memory limit.

5.

60 times.

6.

Any number less than 123

7.

Counters count how many times something happens while an accumulator just keeps a running total. A counter generally only increases by 1, while an accumulator adds whatever value to the total. Ex: counters can count how many times a loop executes or how many students passed a certain test. Accumulators can add up all the marks in a class and give a total, or even add up the prices of items on a shopping list.

8.

```
int sum = 0;
for (int i = 3; i <= 10; i = i + 1){
        sum += i;
}
System.out. println(“Here is the sum: “ + sum)
```

9.

Choosing loop structure depends on if you know how many times the loop should execute and whether or not the loop should only run while the condition is true or if it should run at least once anyway.

11.

a) 10
b) “my ”
c) “my string.”
d) “MY STRING.”
e) “my string.”